

EVALUATING EXPLAINABLE AI METHODS IN DOMAINS OF VARYING HUMAN IMPACT

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Introduction

The research addresses the AI "black box" transparency gap by examining user trust across different risk domains and evaluating Explainable AI (XAI) explanations for clarity, understanding, and trust.

User Study

- **Qualtrics: build** user study to measure SCS & S-TIAS scores.
- **Prolific:** Used to collect survey responses.
- **Survey Integration:** Predictions and XAI explanations were taken from our system.

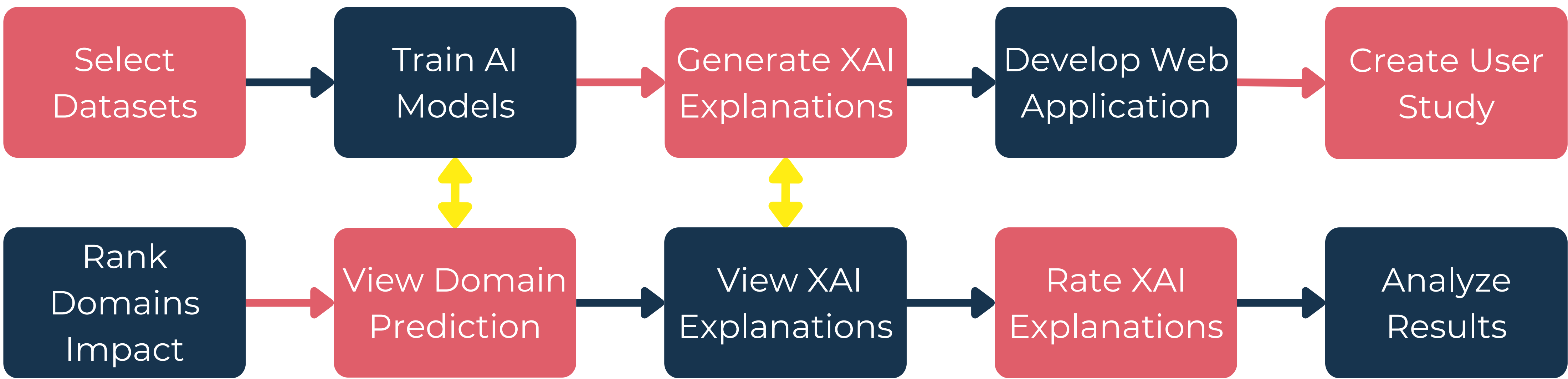
System Methods

- **Domains Selection:** Healthcare, HR, and Text Analysis represent high, medium, and low-risk domains respectfully.
- **AI Models:** MLPClassifier for Healthcare and HR, and Logistic Regression for Text Analysis.
- **XAI Tools:** SHAP - Global Explanations, show feature contributes to the final prediction. LIME - Local Explanations, show feature influence on the specific outcome. DiCE - Counterfactuals, explains the AI's decision by providing "what-if" scenarios.
- **Web Platform:** React web interface to present the predictions and XAI explantions.

Development

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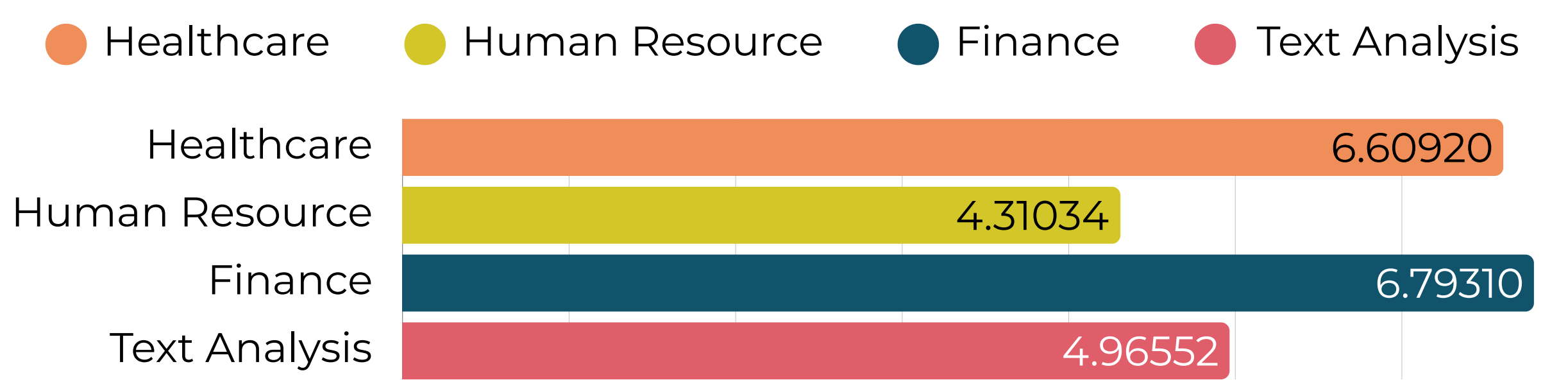
Research



Results

- **Domain Category:** The domain category alone did not significantly affect users' SCS or S-TIAS scores.
- **Explanation Method:** DiCE performed best in Text Analysis, while all methods showed similar results in Healthcare and HR.
- **Understanding and Trust:** A strong positive relationship was found between understanding (SCS) and trust (S-TIAS).

- **Perceived Risk:** Higher perceived risk was linked to slightly higher SCS scores, showing a weak but significant correlation ($r = 0.163$, $p = 0.031$).



Conclusion

- **Context Matters:** XAI explanations should match the domain, data type, and user understanding.
- **No Universal Best Method:** No single method can be assumed to work best across all domains.
- **Perceived Risk:** Influences how users evaluate explanations, as higher risk perception was associated with higher SCS scores.
- **Understanding Builds Trust:** Clear explanations are essential for trust in AI.

Risk Domain	Explanation Method	Mean SCS Score	Mean S-Tias
HR	SHAP	0.608	0.442
	LIME	0.615	0.43
	DiCE	0.655	0.494
	Overall	0.625	0.453
Healthcare	SHAP	0.672	0.531
	LIME	0.644	0.465
	DiCE	0.649	0.498
	Overall	0.654	0.498
Text Analysis	SHAP	0.57	0.481
	LIME	0.618	0.45
	DiCE	0.71	0.575
	Overall	0.635	0.503